DEPARTMENT OF PHYSICS
INDIAN INSTITUTE OF TECHNOLOGY, MADRAS

PH1020 Physics II

Problem Set 2

Electric Potential and Energy in electrostatics Date: 31 Jan 2025

1. A circular sheet of radius a has uniform surface charge density σ. Calculate the potential at a point P on the circumference of the sheet, and also at C, the center of the circle (see Fig. 1).

   Hint: Taking P as the origin and PC as the line corresponding to $\varphi = 0$, the equation of the circle in polar coordinates is:

   $$\varrho = 2a\cos\varphi, \quad -\frac{\pi}{2} \leq \varphi \leq \frac{\pi}{2}.$$

   Take elementary strips of area $2\varrho\varphi\, d\varrho$ as shown in the figure 1 in order to evaluate the potential at P. Choose the limits of integration carefully.

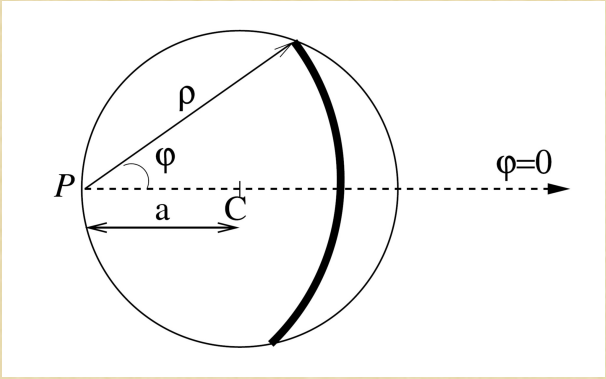


Figure 1: Circular sheet of radius a and uniform surface charge density σ.

2. Two parallel line charges A and B have the same linear charge density λ and the same length L. A extends from $(0,0)$ to $(0,L)$, and B extends from $(a,0)$ to (a,L) in the xy plane. Find the total force on B due to A.

3. The electric field in a certain region in space is given by:

   $$\mathbf{E}(x,y,z) = \frac{k}{\epsilon_0 a^2}\left(x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}\right),$$

   where k and a are constants with appropriate physical dimensions.

   Is the field conservative? If so, calculate the electrostatic potential as well as the charge density at an arbitrary point $P(x,y,z)$.

4. A triangular sheet as shown in the Figure 2 has a uniform charge density σ. Calculate the potential at point P.

5. Calculate the electric potential of a uniformly charged sphere of radius R and charge density ρ_0. Find the work W necessary to create such a sphere.

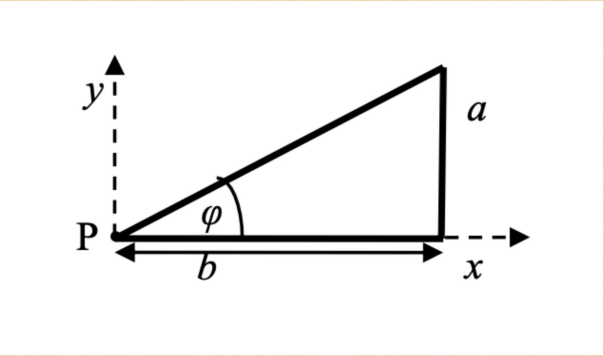


Figure 2: Triangular sheet with uniform charge density σ

6. **Prove the Mean-Value Theorem:** If S_R is the surface of a (mathematical) sphere of radius R whose interior contains no charge, then the potential at the center is equal to the average potential over the surface S_R. Show this by establishing that the average is independent of the radius of the sphere.

7. A spherically symmetric charge distribution (density depends only on the radial coordinate) is placed in a given electric field produced by charges external to the spherical distribution. Prove that the net force experienced by the spherical distribution is as though all of the charge were concentrated at the center of the distribution.
   Hint: Use the mean-value theorem.

1. A circular sheet of radius a has uniform surface charge density σ. Calculate the potential at a point P on the circumference of the sheet, and also at C, the center of the circle (see Fig. 1).

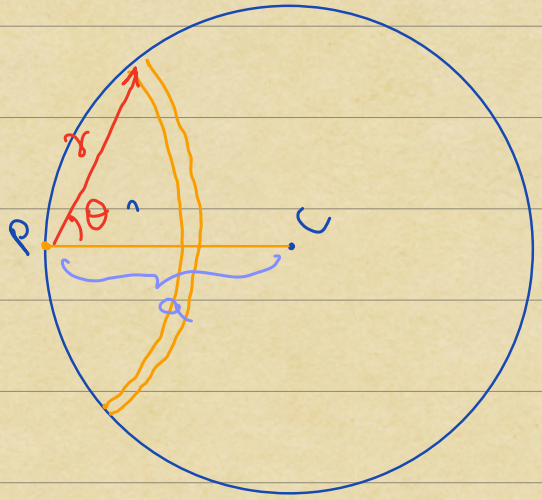
$a^2 = a^2 + r^2 - 2ar\cos\theta$

$\Rightarrow r = 2a\cos\theta$

$dA = 2\pi\theta\, dr$

$\Rightarrow dr = 2a(-\sin\theta\, d\theta)$

$dV = \frac{1}{4\pi\varepsilon_0}\frac{\sigma\, dA}{r}$



$= \frac{1}{4\pi\varepsilon_0}\frac{\sigma \cdot 2\pi\theta\, dr}{r} = \frac{\sigma(2a)}{2\pi\varepsilon_0}(-\theta\sin\theta\, d\theta)$

$V = \frac{\sigma a}{\pi\varepsilon_0}\int_{\pi/2}^{0} -\theta\sin\theta\, d\theta$

$\int_0^{\pi/2} \theta\, d(\cos\theta) = -\int_0^{\pi/2}\cos\theta\, d\theta = -1$

$$\boxed{V = \frac{\sigma a}{\pi\varepsilon_0}}$$

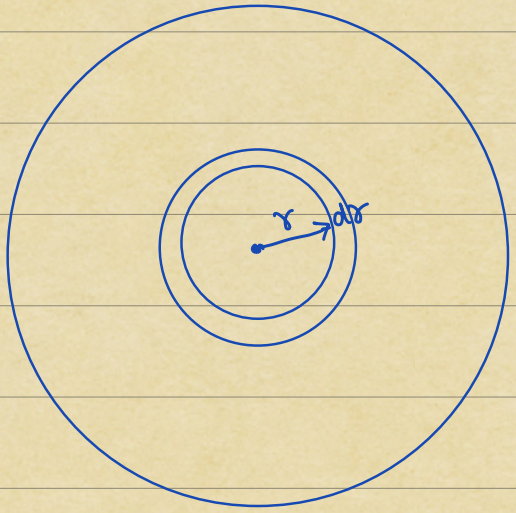
$dq = \sigma\, dA$

$dA = 2\pi r\, dr$

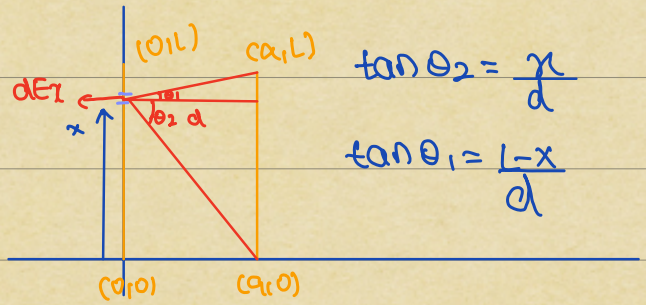
$dV = \frac{1}{4\pi\varepsilon_0}\frac{\sigma \cdot 2\pi r\, dr}{r}$

$dV = \frac{\sigma}{2\varepsilon_0}\int dr$

$$\boxed{V = \frac{\sigma a}{2\varepsilon_0}}$$



2. Two parallel line charges A and B have the same linear charge density λ and the same length L. A extends from $(0,0)$ to $(0,L)$, and B extends from $(a,0)$ to (a,L) in the xy plane. Find the total force on B due to A.



$$\tan\theta_2 = \frac{x}{d}$$

$$\tan\theta_1 = \frac{L-x}{d}$$

$$dE_x = \frac{\lambda}{4\pi\varepsilon_0 d}(\sin\theta_1 + \sin\theta_2)$$

$$dF = dE_x \cdot \lambda dx = \frac{\lambda^2}{4\pi\varepsilon_0 d}(\sin\theta_1 + \sin\theta_2)\cdot dx$$

$$F = \frac{\lambda^2}{d\,4\pi\varepsilon_0}\int_0^L \left(\frac{L-x}{\sqrt{(L-x)^2+d^2}} + \frac{x}{\sqrt{x^2+d^2}}\right) dx$$

$$F = \frac{\lambda^2}{2\pi\varepsilon_0 d}\int_0^L \frac{x}{\sqrt{d^2+x^2}}\,dx$$

$$d^2+x^2 = u^2$$

$$2x\,dx = 2u\,du$$

$$F = \frac{\lambda^2}{2\pi\varepsilon_0 d}\int_d^{\sqrt{L^2+d^2}} \frac{u\,du}{u}$$

$$F = \frac{\lambda^2}{2\pi\varepsilon_0 d}\left(\sqrt{L^2+d^2} - d\right)$$

$$\boxed{F = \frac{\lambda^2}{2\pi\varepsilon_0}\left[\left(\left(\frac{L}{d}\right)^2+1\right)^{1/2} - 1\right]}$$

3. The electric field in a certain region in space is given by:

$$\mathbf{E}(x,y,z) = \frac{k}{\epsilon_0 a^2}\left(x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}\right),$$

where k and a are constants with appropriate physical dimensions.

Is the field conservative? If so, calculate the electrostatic potential as well as the charge density at an arbitrary point $P(x,y,z)$.

$\vec{\nabla} \times \vec{E} = 0$

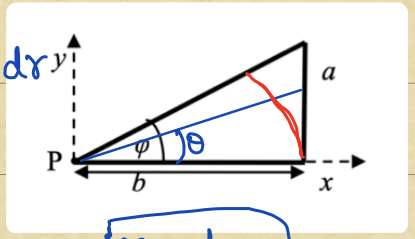
$E = -\nabla \phi$

$\frac{k}{\epsilon_0 a^2}(x\hat{x}+y\hat{y}+z\hat{z}) = -\frac{d\phi}{dx}\hat{x} - \frac{d\phi}{dy}\hat{y} - \frac{d\phi}{dz}\hat{z}$

$\Rightarrow \frac{d\phi}{dx} = \frac{-kx}{\epsilon_0 a^2} \Rightarrow \boxed{\phi = \frac{-k}{2\epsilon_0 a^2}(x^2+y^2+z^2) + C}$

$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} \Rightarrow \boxed{\frac{3k}{a^2} = \rho}$

4. A triangular sheet as shown in the Figure 2 has a uniform charge density σ. Calculate the potential at point P.



$$\int_0^b \frac{1}{4\pi\varepsilon_0}\frac{\sigma \times r\phi\, dr}{r} + \int_b^{b/\cos\phi} \frac{\sigma}{4\pi\varepsilon_0}\frac{r(\phi-\theta)\,dr}{r}$$

$$= \frac{\sigma}{4\pi\varepsilon_0}\phi b + \frac{\sigma}{4\pi\varepsilon_0}\int_0^{\phi} (\phi-\theta)\, b\sec\theta\tan\theta \qquad \boxed{r = \frac{b}{\cos\theta}}$$

$$= \frac{\sigma b}{4\pi\varepsilon_0}\left[\phi + \int_0^{\phi}(\phi-\theta)\sec\theta\tan\theta\, d\theta\right]$$

$$= \frac{\sigma b}{4\pi\varepsilon_0}\left[\phi + \phi\sec\phi - \int_0^{\phi}\theta\, d(\sec\theta)\, d\theta\right]$$

$\theta\sec\theta - \ln|\sec\theta+\tan\theta|$

$-\phi\sec\phi + \ln|\sec\phi+\tan\phi|$

$$= \frac{\sigma b}{4\pi\varepsilon_0}\left[\ln|\sec\phi+\tan\phi|\right]$$

we can take line charges too!

5. Calculate the electric potential of a uniformly charged sphere of radius R and charge density ρ_0. Find the work W necessary to create such a sphere.

$$V = -\int E\,dr$$

$$= -\int_{\infty}^{R} \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\,dr - \int_{R}^{r} \frac{Qr}{4\pi\varepsilon_0}\frac{dr}{R^3}$$

$$= \frac{\rho_0 \times 4/3\,\pi R^3}{4\pi\varepsilon_0 R} + \frac{4/3\,\pi R^3 \rho_0}{4\pi\varepsilon_0 R^3}\left[\frac{R^2 - r^2}{2}\right]$$

$$= \frac{\rho_0 R^2}{3\varepsilon_0} + \frac{\rho_0[R^2 - r^2]}{6\varepsilon_0} = \boxed{\frac{\rho_0[3R^2 - r^2]}{6\varepsilon_0}}$$

$r \to 0 = \rho_0 R^2/2\varepsilon_0$

$r \to R \Rightarrow \quad V = \frac{\rho_0 R^2}{3\varepsilon_0}$.

$dW = dq \cdot V$

$dq = 4\pi r^2 dr\,\rho_0 \Rightarrow \int_0^R 4\pi r^2 dr\,\rho_0 \times \frac{\rho_0 r^2}{3\varepsilon_0}$

$= \frac{4\pi\rho_0^2}{3\varepsilon_0}\frac{R^5}{5}$

$$\boxed{V_{sph} = \frac{4\pi}{15}\rho_0^2 R^5}$$

6. **Prove the Mean-Value Theorem:** If S_R is the surface of a (mathematical) sphere of radius R whose interior contains no charge, then the potential at the center is equal to the average potential over the surface S_R. Show this by establishing that the average is independent of the radius of the sphere.

Let center be at $(x_0, y_0, z_0) = \gamma_0$

$$\gamma(R,\theta,\phi) = (x_0 + R\sin\theta\cos\phi,\ y_0 + R\sin\theta\sin\phi,\ z_0 + R\cos\theta)$$

$$\phi_{avg} = \frac{1}{4\pi R^2}\int dS\ \phi(\gamma)$$

$$= \frac{1}{4\pi}\int \sin\theta\, d\theta\, d\phi\ \phi(\gamma)$$

$$\frac{d}{dR}\phi(\gamma) = \nabla\phi(\gamma)\cdot\hat{\gamma} \Rightarrow \frac{d\phi_{avg}}{dR} = \frac{1}{4\pi}\int d\Omega\ \underbrace{\nabla\phi(\gamma)\cdot\hat{\gamma}}_{\text{flux inside the sphere} = 0} = 0$$

$\Rightarrow \phi(\gamma)$ doesn't depend on Radius of sphere

$$\Rightarrow \phi_{avg} = \frac{1}{4\pi}\int d\Omega\ \phi(\gamma_0) = \phi(\gamma_0)$$

7. A spherically symmetric charge distribution (density depends only on the radial coordinate) is placed in a given electric field produced by charges external to the spherical distribution. Prove that the net force experienced by the spherical distribution is as though all of the charge were concentrated at the center of the distribution.

$$U(\gamma) = \int \rho(\gamma)\cdot\gamma^2 d\Omega\cdot\phi(\gamma_0+\gamma)$$

$$= \left(\int \rho(\gamma)\cdot\gamma^2 d\Omega\right)\phi(\gamma_0)$$

$$= Q\cdot\phi(\gamma_0)$$

$$\vec{F} = -\nabla U(\gamma) = -\nabla(Q\cdot\phi(\gamma_0))$$

$$= -Q\,\nabla(\phi(\gamma_0)) = Q\vec{E}(\gamma_0)$$

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Problem Set 2 - Solutions

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1. Problem-1

Question: A circular sheet of radius a has uniform surface charge density σ. Calculate the potential at a point P on the circumference of the sheet, and also at C, the center of the circle (see Fig.1).

Hint: Taking P as the origin and PC as the line corresponding to $\varphi = 0$, the equation of the circle in polar coordinates is:

$$\varrho = 2a\cos\varphi, \quad -\frac{\pi}{2} \leq \varphi \leq \frac{\pi}{2}.$$

Take elementary strips of area $2\varrho\varphi\, d\varrho$ as shown in the figure in order to evaluate the potential at P. Choose the limits of integration carefully.

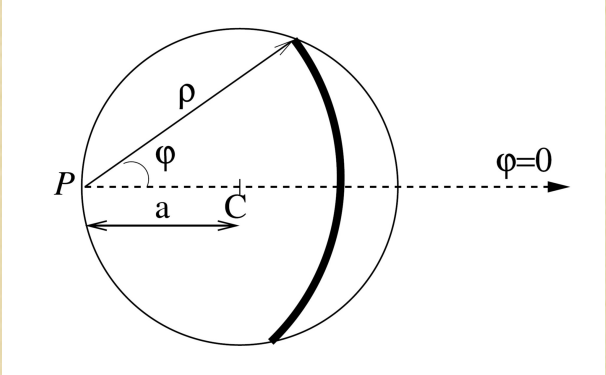


Figure 1: Circular sheet of radius a and uniform surface charge density σ.

Solution:

(a) Potential at P : We wish to compute the potential at a point P on the circumference of the sheet. The general expression for the potential at a point $\mathbf{r}$ due to the surface charge distribution is (the coordinate $\mathbf{r}'$ runs over the surface charge distribution)

$$\phi(\mathbf{r}) = k \int dS' \frac{\sigma(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} \tag{1}$$

Using the coordinate system suggested in the hint, i.e., (ϱ', φ') with $-\pi/2 \leq \varphi' \leq \pi/2$ and for a given φ', one has $0 \leq \varrho' \leq 2a\cos\varphi'$. Thus, we have

$$\int dS' = \int_{-\pi/2}^{\pi/2} d\varphi' \int_0^{2a\cos\varphi'} d\varrho'\, \varrho'$$

It is useful to check that the limits are fine by just carrying out the integral with integrand equal 1 gives the area to be πa^2. In this coordinate system, the point P has $\mathbf{r} = 0$. Using $|\mathbf{r} - \mathbf{r}'| = |\mathbf{r}'| = \varrho'$, we obtain

$$\begin{aligned}\phi(P) &= k\int_{-\pi/2}^{\pi/2} d\varphi' \int_0^{2a\cos\varphi'} d\varrho'\,\varrho' \frac{\sigma}{\varrho'} = k\sigma \int_{-\pi/2}^{\pi/2} d\varphi' \int_0^{2a\cos\varphi'} d\varrho' \\ &= (2ak\sigma)\int_{-\pi/2}^{\pi/2} d\varphi' \cos\varphi' \\ \phi(P) &= (4ak\sigma).\end{aligned}$$

Alternate method: A second way of doing problem 1 is to exchange the order of integration of (ϱ, φ). (Below $\arccos(x) = \cos^{-1}(x)$)

$$\int dS' = \int_0^{2a} d\varrho'\,\varrho' \int_{-\arccos(\varrho'/(2a))}^{+\arccos(\varrho'/(2a)} d\varphi$$

Then we can repeat the computation for $\phi(P)$

$$\begin{aligned}\phi(P) = \int dS' \frac{k\sigma}{\varrho'} &= k\sigma \int_0^{2a} d\varrho \int_{-\arccos(\varrho'/(2a))}^{+\arccos(\varrho'/(2a)} d\varphi \\ &= k\sigma \int_0^{2a} d\varrho' \times 2\arccos\left(\frac{\varrho'}{2a}\right) \\ &= (4ak\sigma)\int_0^1 dw \arccos w \quad (\text{ defining } w = \varrho'/2a) \\ &= (4ak\sigma)\end{aligned}$$

It can be shown that the integral $I = \int_0^1 dw \arccos w = 1$ for instance, by integrating by parts to obtain $I = \int_0^1 \frac{w dw}{\sqrt{1-w^2}}$.

(b) Potential at C : Now it makes sense to work with a coordinate system centered at C. Again $|\mathbf{r} - \mathbf{r}'| = |\mathbf{r}'| = \varrho'$. The limits of the area integral are suitably adjusted to this coordinate system.

$$\begin{aligned}\phi(C) &= k\int_0^{2\pi} d\varphi' \int_0^a d\varrho'\,\varrho' \frac{\sigma}{\varrho'} = k\sigma \int_0^{2\pi} d\varphi' \int_0^a d\varrho' \\ \phi(C) &= (2\pi ak\sigma).\end{aligned}$$

2. Problem-2

Question: Two parallel line charges A and B have the same linear charge density λ and the same length L. A extends from $(0, 0)$ to $(0, L)$, and B extends from $(a, 0)$ to (a, L) in the xy plane. Find the total force on B due to A.

Solution:

Consider two charge elements dy, and dy'. The force on dy' due to dy is given as, (see figure below)

$$dF = \frac{\lambda^2 dy dy'}{4\pi\epsilon_0 \left(a^2 + (y' - y)^2\right)^{3/2}} \left[a\hat{\mathbf{x}} + (y' - y)\,\hat{\mathbf{y}}\right].$$

By symmetry the force in the y direction vanishes. This can be checked by interchanging y and y' resulting in the same magnitude of force in the y-direction but with a reversal in sign. Thus we evaluate

$$F_x = dF_x = \frac{\lambda^2 a}{4\pi\epsilon_0}\int_0^L\int_0^L \frac{dydy'}{\left(a^2+(y'-y)^2\right)^{3/2}}$$

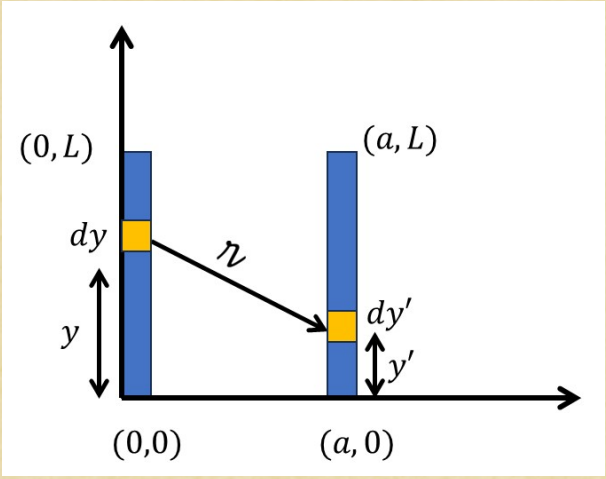


Figure 2: Two parallel line charges A and B have the same linear charge density λ and the same length L. (The figure depicts a line charge. The apparent width is for illustrative purposes only; a line charge has no width.)

We use

$$\int \frac{dx}{(x^2+a^2)^{3/2}} = \frac{x}{a^2\sqrt{x^2+a^2}}$$

, to solve for the force above, and we obtain on doing the y integral

$$F_x = \frac{\lambda^2 a}{4\pi\epsilon_0 a^2}\left[\int_0^L \frac{(L-y')\,dy'}{\sqrt{(L-y')^2+a^2}} + \int_0^L \frac{y'dy'}{\sqrt{y'^2+a^2}}\right]$$

Doing the y' integral gives

$$F_x = \frac{\lambda^2 2a}{4\pi\epsilon_0 a}\left[\sqrt{1+L^2/a^2}-1\right].$$

Thus the force is given by

$$\mathbf{F} = \frac{\lambda^2}{2\pi\epsilon_0}\left[\sqrt{1+L^2/a^2}-1\right]\hat{\mathbf{x}}$$

3. Problem-3

Question: The electric field in a certain region in space is given by:

$$\mathbf{E}(x,y,z) = \frac{k}{\epsilon_0 a^2}\left(x\hat{\mathbf{x}}+y\hat{\mathbf{y}}+z\hat{\mathbf{z}}\right),$$

where k and a are constants with appropriate physical dimensions.

Is the field conservative? If so, calculate the electrostatic potential as well as the charge density at an arbitrary point $P(x, y, z)$.

Solution:

Let us look at

$$\vec{E} = \frac{k}{\epsilon_0 a^2}(x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}})$$

which can be obviously written as

$$\vec{E} = \frac{k}{\epsilon_0 a^2}\vec{r}$$

From what we have learned from PH1010, for such forces $\vec{\nabla} \times \vec{r} = 0$, which implies a conservative field. To compute the potential, we have

$$\Phi = -\int \vec{E} \cdot \vec{d\ell} = -\frac{k}{\epsilon_0 a^2}\int r dr + \mathrm{C} = -\frac{kr^2}{2\epsilon_0 a^2} + \mathrm{C}$$

To find ρ use Gauss's law

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0} = \frac{k}{\epsilon_0 a^2}\vec{\nabla} \cdot \vec{r} = \frac{3k}{\epsilon_0 a^2}$$

Thus,

$$\rho = \frac{3k}{a^2}$$

4. Problem-4

Question: A triangular sheet as shown in the figure 3 has a uniform charge density σ. Calculate the potential at point P.

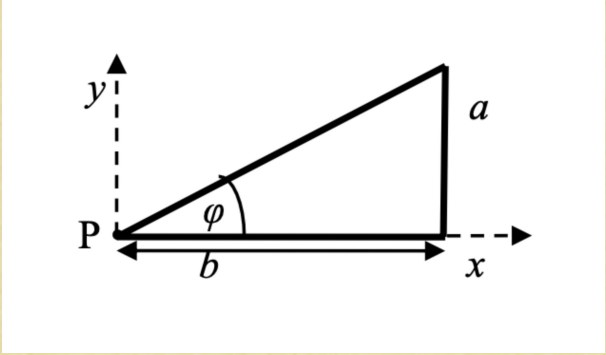


Figure 3: Triangular sheet with uniform charge density σ

Solution:

The potential due to the area element "$dxdy$" at the point P is given as

$$dV = \frac{\sigma dxdy}{4\pi\epsilon_0\sqrt{x^2 + y^2}},$$

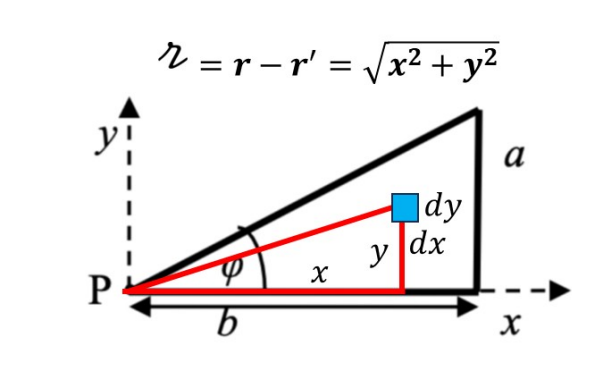


Figure 4: The potential at the point P due to the area element "$dxdy$"

with $y = x\tan\phi$. Integrating the above we obtain,

$$V = \int dV = \frac{\sigma}{4\pi\epsilon_0}\int_0^b \int_0^{x\tan\phi} \frac{dxdy}{\sqrt{x^2+y^2}} \tag{2}$$

The above integral can be solved by putting $y = x\tan t$. This implies that

$$\int \frac{dy}{\sqrt{x^2+y^2}} = \int \sec t$$

Now, consider

$$\sec t dt = \int \sec t \frac{\sec t + \tan t}{\sec t + \tan t} dt$$

One can easily check that the above integral is of the form $\frac{du}{u}$, which can be evaluated to give $\ln|u|$. Thus, re-writing in terms of the variable t, we get

$$\int \sec t = \ln|\sec t + \tan t|$$

Using this result in Eq. 2, we get

$$V = \frac{\sigma}{4\pi\epsilon_0}\int_0^b dx \ln(\tan\phi + \sec\phi) = \frac{\sigma b}{4\pi\epsilon_0}\ln(\tan\phi + \sec\phi)$$

Now, substituting $\tan\phi = \frac{a}{b}$ and $\sec\phi = \frac{\sqrt{a^2+b^2}}{b}$ in the above gives

$$V = \frac{\sigma b}{4\pi\epsilon_0}\ln\left(\frac{a+\sqrt{a^2+b^2}}{b}\right)$$

5. Problem-5

Question: Calculate the electric potential of a uniformly charged sphere of radius R and charge density ρ_0. Find the work W necessary to create such a sphere.
Solution:

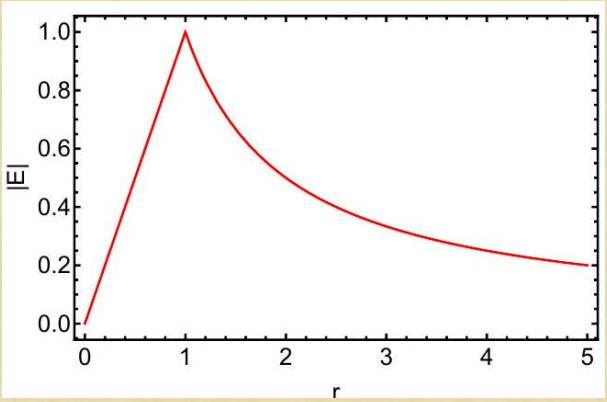


Figure 5: Plot of the magnitude of the electric field as a function of radial distance of uniformly charged solid sphere. (**Note:** r is in the units of R and $|\mathbf{E}|$ is in the units of $\frac{\rho_0 R}{3\epsilon_0}$)

We know the electrostatic potential $V(r) = -\int_\infty^r \mathbf{E} \cdot d\mathbf{l}$. So let's find the electric field first.

For $r < R$: Applying Gauss's law

$$\oint \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{\text{enc}}}{\epsilon_0}$$
$$|E| 4\pi r^2 = \rho_0 \frac{4}{3}\pi r^3 \frac{1}{\epsilon_0}$$
$$\mathbf{E} = \frac{\rho_0}{3\epsilon_0} r\hat{\mathbf{r}}$$

For $r > R$: Applying Gauss's law again

$$\oint \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{\text{enc}}}{\epsilon_0}$$
$$|E| 4\pi r^2 = \rho_0 \frac{4}{3}\pi R^3 \frac{1}{\epsilon_0}$$
$$\mathbf{E} = \frac{\rho_0 R^3}{3\epsilon_0} \frac{1}{r^2}\hat{\mathbf{r}}$$

Now let's find the potential. For $r > R$:

$$\begin{aligned} V(r) &= -\int_\infty^r \mathbf{E} \cdot d\mathbf{l} \\ &= -\int_\infty^r \frac{\rho_0 R^3}{3\epsilon_0} \frac{1}{r^2} dr \\ &= \frac{\rho_0 R^3}{3\epsilon_0} \frac{1}{r}\Big|_\infty^r \\ &= \frac{\rho_0 R^3}{3\epsilon_0} \frac{1}{r} \end{aligned}$$

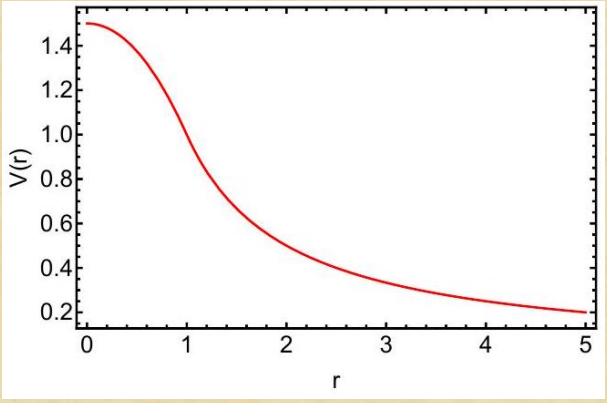


Figure 6: Plot of the potential as a function of radial distance of uniformly charged solid sphere. (**Note:** r is in the units of R and $V(r)$ is in the units of $\frac{\rho_0 R^2}{3\epsilon_0}$)

For $r < R$:

$$\begin{aligned} V(r) &= -\int_{\infty}^{r} \mathbf{E} \cdot d\mathbf{l} \\ &= -\int_{\infty}^{R} \frac{\rho_0 R^3}{3\epsilon_0}\frac{1}{r^2}dr - \int_{R}^{r} \frac{\rho_0}{3\epsilon_0} r dr \\ &= \frac{\rho_0 R^2}{3\epsilon_0} - \left(\frac{\rho_0}{3\epsilon_0}\frac{r^2}{2}\right)\Big|_{R}^{r} \\ &= \frac{\rho_0 R^2}{6\epsilon_0}\left(3 - \frac{r^2}{R^2}\right) \end{aligned}$$

Work done to create such a sphere: Following the equation (2.43) of Griffith's book

$$\begin{aligned} W &= \frac{1}{2}\int \rho V d\tau \\ &= \frac{1}{2}\frac{\rho_0^2 R^2}{6\epsilon_0}\int_0^{2\pi} d\phi \int_0^{\pi} \sin\theta d\theta \int_0^{R} \left(3 - \frac{r^2}{R^2}\right) r^2 dr \\ &= \frac{1}{2}\frac{\rho_0^2 R^2}{6\epsilon_0} 4\pi \left[r^3 - \frac{r^5}{5R^2}\right]_0^{R} \\ &= \frac{1}{2}\frac{\rho_0^2 R^2}{6\epsilon_0} 4\pi \frac{4R^3}{5} \\ &= \frac{4\pi}{15\epsilon_0}\rho_0^2 R^5 \end{aligned}$$

Also, an alternative method

$$\begin{aligned} W &= \frac{1}{2}\epsilon_0 \int E^2 d\tau \\ &= \frac{\epsilon_0}{2}\int_0^{R} \frac{\rho_0^2 r^2}{9\epsilon_0^2} 4\pi r^2 dr + \frac{\epsilon_0}{2}\int_R^{\infty} \frac{\rho_0^2 R^6}{9\epsilon_0^2 r^4} 4\pi r^2 dr \\ &= \frac{4\pi}{15\epsilon_0}\rho_0^2 R^5 \end{aligned}$$

6. Problem-6

Question: Prove the Mean-Value Theorem: If S_R is the surface of a (mathematical) sphere of radius R whose interior contains no charge, then the potential at the center is equal to the average potential over the surface S_R. Show this by establishing that the average is independent of the radius of the sphere.

Solution:

Let the sphere S_R be centered at the point $\mathbf{r}_0 = (x_0, y_0, z_0)$. Let

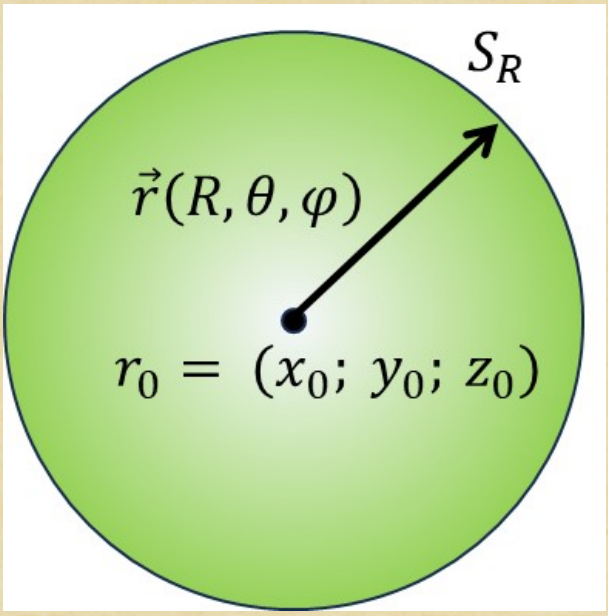


Figure 7: Sphere S_R centered at the point $\mathbf{r}_0 = (x_0, y_0, z_0)$.

$$\mathbf{r}(R, \theta, \varphi) = (x_0 + R\sin\theta\cos\varphi)\,\hat{\mathbf{x}} + (y_0 + R\sin\theta\sin\varphi)\,\hat{\mathbf{y}} + (z_0 + R\cos\theta)\,\hat{\mathbf{z}}.$$

Then, as we vary (θ, φ), the vector $\mathbf{r}(R, \theta, \varphi)$ we obtain the surface of the sphere of radius r centered at (x_0, y_0, z_0).

The average potential, $\langle\phi\rangle$, which is a function of R, is given by

$$\begin{aligned}\langle\phi\rangle &= \frac{1}{4\pi R^2}\int dS\phi(\mathbf{r}(R, \theta, \varphi)) \\ &= \frac{1}{4\pi}\int d\Omega\phi(\mathbf{r}(R, \theta, \varphi))\end{aligned}$$

where the R^2 gets cancelled by the factor of R^2 in dS. In order to study the R dependence of $\langle\phi\rangle$, we take the R derivative to obtain. [1]

[1] **Note :** We need to use the identity

$$\frac{\partial}{\partial R}(\phi(\mathbf{r}(R, \theta, \varphi)) = \boldsymbol{\nabla}\phi(\mathbf{r}) \cdot \frac{\partial \mathbf{r}}{\partial R} = \boldsymbol{\nabla}\phi(\mathbf{r}) \cdot \hat{\mathbf{r}}.$$

$$\begin{aligned}\frac{d}{dR}\langle\phi\rangle &= \left(\frac{1}{4\pi}\int d\Omega \frac{\partial}{\partial R}\phi(\mathbf{r}(R,\theta,\varphi))\right)\\ &= \frac{1}{4\pi}\int d\Omega\ \hat{\mathbf{r}}\cdot\boldsymbol{\nabla}(\phi) \text{ which is the electric flux up to numerical factors}\\ &= 0 \quad \text{for a charge-free region.}\end{aligned}$$

Thus, we see that $\langle\phi\rangle$ is a constant. The value of the constant is fixed by observing that the limit $R \to 0$ is nice and we get

$$\langle\phi\rangle = \lim_{R\to 0}\frac{1}{4\pi}\int d\Omega\phi(\mathbf{r}(R,\theta,\varphi)) = \phi\left(\mathbf{r}_0\right) \times \frac{1}{4\pi}\int d\Omega = \phi\left(\mathbf{r}_0\right)$$

$$completing the required proof.$$

7. Problem-7

Question: A spherically symmetric charge distribution (density depends only on the radial coordinate) is placed in a given electric field produced by charges external to the spherical distribution. Prove that the net force experienced by the spherical distribution is as though all of the charge were concentrated at the center of the distribution.
Hint: Use the mean-value theorem.
Solution:

We compute the force by first computing the potential energy of the spherically symmetric charge distribution. Let $\phi(\mathbf{r})$ denote the potential due to charges external to the spherically symmetric charge distribution. Let the centre of the spherical symmetry be at the point $\mathbf{r}_0$. Then, we obtain that the potential energy is

$$\begin{aligned}U\left(\mathbf{r}_0\right) &= \int dV\rho(\mathbf{r})\phi\left(\mathbf{r}_0+\mathbf{r}\right)\\ &= \int drr^2\rho(r)\int d\Omega\phi\left(\mathbf{r}_0+\mathbf{r}\right)\\ &= \int drr^2\rho(r)4\pi\phi\left(\mathbf{r}_0\right)\end{aligned}$$

where we use the mean-value theorem to obtain the last line. Thus, we see that $U\left(\mathbf{r}_0\right) = Q\phi\left(\mathbf{r}_0\right)$ where Q is the total charge of the spherical distribution. The net force, $\mathbf{F}$, on the spherically symmetric charge distribution is then

$$\mathbf{F} = -\nabla_0 U\left(\mathbf{r}_0\right) = -Q\nabla_0\phi\left(\mathbf{r}_0\right) = Q\mathbf{E}\left(\mathbf{r}_0\right)$$

where $\boldsymbol{\nabla}_0$ is the gradient for the coordinate $\mathbf{r}_0$. This proves the statement.
[2]

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